Unveiling the biodiversity and conservation significance of Jianfengling a camera-trapping survey of mammals in Hainan Tropical Rainforest National Park

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Abstract

Background

Jianfengling on southwestern Hainan Island is sanctuary for a diverse range of wild animals. However, the exact extent of mammal species diversity and conservation status in Jianfengling remains largely unknown, despite their crucial role in maintaining the ecological balance. This study focused on the diversity and distribution status of mammal species in Jianfengling, Hainan Tropical Rainforest National Park, China.

Results

The survey, which spans from October 2020 to November 2021, with 41,571 camera days and 8,091 independent detections, revealed 15 mammalian species belonging to 6 orders and 10 families. Among these, one was categorized as Critically Endangered, one as Near Threatened on the IUCN Red List, two were categorized as Critically Endangered or Endangered on the Red List of China's Vertebrates, and five were China's national first-class or second-class key protected wildlife. Notably, populations of the Chinese pangolin (*Manis pentadactyla*) were confirmed to persist in the wild of Jianfengling. In terms of the relative abundance indices (RAIs) of mammals captured by camera traps, the most prevalent species identified was Asiatic brush-tailed porcupine (*Atherurus macrourus*), followed by the wild boar (*Sus scrofa*), pallas's squirrel (*Callosciurus erythraeus*), and Hainan muntjac (*Muntiacus nigripes*). The monitoring also captured a significant number of domestic dogs, as well as human disturbances.

Conclusion

These findings underscore the importance of conserving these mammals and emphasize the need for conservation efforts to protect their habitats and reduce human activities that threaten their survival, thereby maintaining the ecological balance of the region.

Introduction

Tropical forests are habitats for diverse species of endemic flora and fauna [1]. Despite covering just 6% of the Earth's land surface, tropical forests play a pivotal role as habitats for even 90% of the total number of species worldwide [2]. This circumstance designates these areas as biodiversity hotspots, underscoring the imperative to conserve and safeguard these natural habitats [3]. However, tropical forests are at risk from increasing human activity [1]. Therefore, protecting tropical forests is currently the most pressing issue of concern worldwide.

Mammals are a crucial component of forest ecosystems and play a significant role in ecosystem services and functions such as nutrient cycling, plant growth, and seed dispersal owing to their life cycles and reproductive strategies [4–7]. However, they are also among the most threatened animal groups, facing issues such as habitat loss, overexploitation, climate change, and invasive species [8]. Therefore, understanding the biodiversity, distribution, abundance, and conservation status of mammals is essential for their effective management and protection.

Owing to their dense vegetation and complex community structures, tropical rainforests provide suitable microhabitats for various mammals, but their visibility in tropical rainforests is limited [9]. Many mammal populations have low densities, secretive behaviors, and nocturnal activity patterns in tropical rainforests [10]. Therefore, one of the primary challenges in researching and protecting tropical rainforest mammals lies in the considerable difficulty associated with detecting and monitoring them. Camera-trapping technology, a method that offers advantages such as long-term, continuous, covert, and minimally disruptive monitoring, is an economical, efficient, and easy-to-use "noninvasive" tool suitable for studying forest wildlife that is rare, secretive, and nocturnal [11–13]. This technology aids in the study of mammal diversity, distribution, and activity patterns [14–18].

The historical coverage rate of primeval forests was greater than 90% on Hainan Island; this rate decreased to 25.5% in 1956. Owing to urbanization, logging economic activities, and the rapid development of tropical agriculture, tropical rainforests have experienced rapid loss and degradation, with the area reaching only 4% of Hainan Island by the end of the 20th century [19]. Located southwest Hainan Island, Jianfengling stands as one of the best-preserved tropical forest areas on the island, with rich biodiversity, earning it the nickname "the gene pool of biological species on the northern edge of the tropics" [20]. To protect the tropical rainforest ecosystem and its biodiversity in Jianfengling, China established the Hainan Jianfengling Nature Reserve in 1956, and in 2020, the Jianfengling Branch of Hainan Tropical Rainforest National Park was formally inaugurated. Historical records indicate that Jianfengling is home to 68 species of mammals, including the clouded leopard (*Neofelis nebulosa*), the Hainan gibbon (*Nomascus hainanus*), Eld's deer (*Rucervus eldii*), and the Chinese pangolin (*Manis pentadactyla*) [21]. However, the exact extent of mammal species diversity and conservation status in Jianfengling remains largely unknown due to the lack of baseline data necessary for systematic monitoring, surveys, and management, and there is no systematic research or reliable estimates of their density and abundance available. In recent years, increasing anthropogenic disturbances, including tropical crop implantation, ecotourism development, and even poaching activities, have exacerbated habitat loss and fragmentation in local wildlife, potentially threatening the survival of certain species [22, 23]. Therefore, identifying mammal species in Jianfengling and implementing protective measures are essential for preserving the ecological integrity and cultural heritage of the region.

In this study, we concentrated on researching the diversity and distribution status of mammal species in Jianfengling, Hainan Tropical Rainforest National Park, China. We conducted a camera-trapping survey and statistical analysis to evaluate the importance of their conservation. Our objectives included compiling an inventory of species and providing an initial assessment of their relative abundance to understand the potential conservation importance of Jianfengling. Additionally, we assessed the preliminary threats and challenges to mammal conservation in Jianfengling and proposed recommendations for future actions.

Methods

Study Area

Jianfengling, located southwest of Hainan Island, is located within the Jianfengling Branch of Hainan Tropical Rainforest National Park (108°45'-109°03' E, 18°38'-18°52' N; Fig. 1a). This territory is predominantly populated by the Li ethnic group. Its area is 678 km^2, with an elevation range of 112–1412 m (Fig. 1b). It is located on the northern edge of the tropics and has a tropical monsoon climate and distinct dry and rainy seasons. The annual average rainfall is 2265.8 mm, with the annual precipitation concentrated between May and October. The annual average temperature is 24.5°C, with rainfall and heat occurring over the same period. Its unique ecological environment supports a variety of vegetation types, mainly comprising tropical montane rainforest, tropical evergreen monsoon forest, tropical valley rainforest, and tropical semideciduous monsoon forest, accounting for 45.96%, 46.53%, 1.58% and 5.77% of the total area, respectively.

Camera Trapping

Camera-trapping surveys were conducted from October 2020 to November 2021 to monitor mammals in Jianfengling via 123 cameras (E3H series, Shenzhen Ereagle Technology Co. Ltd., Shenzhen, China; Ltl 6511 series, Zhuhai Ltl Acorn Electronics Co. Ltd., Zhuhai, China). Each camera was configured to operate continuously for 24 h. Each event was registered by 3 photographs followed by a 10–20 s video, with a 1 s interval before the next trigger. Owing to the shooting effect being easily affected by factors such as wind and changes in light intensity, the sensitivity of passive infrared detection was registered to be moderate. The cameras were mounted 0.5–1.0 m above ground level, and the camera lenses were adjusted to be parallel to the ground, with a broad view and avoiding direct sunlight. Camera traps were strategically positioned in areas of high wildlife activity, animal trails, ridge tops and fallen logs. The location of each camera trap station was recorded via a global positioning system (GPS), and the mean distance between neighboring cameras was approximately 500 m. Sixty-six, fifty, six and one camera traps were distributed in tropical montane rainforests, tropical evergreen monsoon forests, tropical semideciduous monsoon forests, and tropical valley rainforests, respectively (Fig. 1c). The heights of these camera traps ranged from 237 to 1,098 m. These cameras were checked for working conditions every 3 months.

Data analysis

The method of data processing used involved identifying mammalian species in the images, analyzing the data to estimate the frequency of animal occurrence, and combining a geographic information system (GIS) to analyze the distribution of these species. The identification of mammalian species in the images followed the criteria of Smith and Xie [24], Jiang et al. [25], and Wei et al. [26].

The conservation status of mammals was assessed at both the global and national levels according to the threat categories assigned to the IUCN Red List (http://www.iucnredlist.org/) and the Red List of China's vertebrates [27]. The survey effort was quantified by the total number of camera-days, where every 24 h of continuous operation of an infrared camera in the field counts as one camera-day. A single capture event was defined as any sequence for a given species taken more than 30 min after the previous sequence of that species at the same location [28–30]. Additionally, we tallied the total number of independent capture events for each species and determined the grid occupancy (GO) that recorded each species [29, 31]. The relative abundance indices (RAIs) were also calculated according to the methods suggested by O'Brien et al. [11] as a species quantitative index to assess the abundance of species within the territory. The GO and RAI values were calculated via the following formulae:

$$GO = \frac{S}{T} \times 100\%; RAI = \frac{Ai}{N} \times 1000$$

where S is the number of camera traps that recorded the species; T represents the total number of camera traps; Ai represents the number of independent mammal photographs of the i-th species; and N is the total number of camera days.

Results

Mammalian Diversity in Jianfengling

Among the 123 camera traps, 9 failed to generate usable data due to loss. Over 41,571 camera days, a total of 151,192 photos and videos were captured. There were 8,091 independent detections for mammalian species, of which 832 were rodent images of unspecified species and 141 were people and domestic dogs. In total, 15 species of mammals belonging to 6 orders and 10 families were identified (Table 1). Among these detected species, one was categorized as Critically Endangered, and one as Near Threatened on the IUCN Red List. According to the Red List of China's Vertebrates, one was categorized as Critically Endangered, one as Endangered, three as Vulnerable, and three as Near Threatened. Additionally, one was designated China's national first-class key for protected wildlife, and four were designated China's national second-class key for protected wildlife (Table 1). Notably, we discovered four species that are rare in China: the Chinese pangolin, the common palm civet (*Paradoxurus hermaphrodites*), the Hainan muntjac (*Muntiacus nigripes*), and the black giant squirrel (*Ratufa bicolor*) (Fig. 2).

Table 1
Mammalian species were monitored at the Jianfengling Branch of Hainan Tropical Rainforest National Park, China, from October 2020 to November 2021.

Species (by Order)	**IUCN Red List Status** [a]	**China's Vertebrate Red List Status** [b]	**China's National Protected Category** [c]	**Elevation Range Observed (m)**	**Habitat types** [d]	**Total No. of Independent Capture Events**	**Grid Occupancy (GO)**	**Relative Abundance Index (RAI)**
Primates								
Cercopithecidae								
Rhesus monkey *Macaca mulatta*	LC	LC	[illegible]	270–1077	E,M,S,V	84	0.36	2.02
Pholidota								
Manidae								
Chinese pangolin *Manis pentadactyla*	CR	CR	[illegible]	454–974	E,M	12	0.08	0.29
Carnivora								
Viverridae								
Common palm civet *Paradoxurus hermaphroditus*	LC	EN	[illegible]	237–1077	E,M,S	177	0.42	4.26
Felidae								
Mainland leopard cat *Prionailurus bengalensis*	LC	VU	[illegible]	237–1077	E,M	69	0.31	1.66
Mustelidae								
Yellow-bellied weasel *Mustela kathiah*	LC	NT	/	464	E	1	0.01	0.02
Small-toothed ferret badger *Melogale moschata*	LC	NT	/	269–1077	E,M,S	604	0.52	14.53
Artiodactyla								
Cervidae								
Hainan muntjac *Muntiacus nigripes*	NE	VU	[illegible]	310–1077	E,M,S	1027	0.50	24.70
Suidae								
Wild boar *Sus scrofa*	LC	LC	/	237–1077	E,M,S,V	1148	0.82	27.62

[a,b] CR, Critically Endangered; EN, Endangered; VU, Vulnerable; NT, Near Threatened; LC, Least Conscern; NE, Not Evaluated; [c] [illegible], China national first-class key protected wildlife; [illegible], China national second-class key protected wildlife; [d] E, tropical evergreen monsoon forest; M, tropical montane rainforest; S, tropical semdeciduous monsoon forest; V, tropical valley rainforest

Species (by Order)	IUCN Red List Status [a]	China's Vertebrate Red List Status [b]	China's National Protected Category [c]	Elevation Range Observed (m)	Habitat types [d]	Total No. of Independent Capture Events	Grid Occupancy (GO)	Relative Abundance Index (RAI)
Scandentia								
Tupaiidae								
Northern treeshrew *Tupaia belangeri*	LC	LC	/	269–956	E,M,S	52	0.18	1.25
Rodentia								
Hystricidae								
Malayan porcupine *Hystrix brachyura*	LC	LC	/	441–683	E,M	11	0.05	0.26
Asiatic brush-tailed porcupine *Atherurus macrourus*	LC	LC	/	237–1098	E,M,S	1838	0.58	44.21
Sciuridae								
Black giant squirrel *Ratufa bicolor*	NT	VU	[illegible]	564–883	E,M	2	0.02	0.05
Pallas's squirrel *Callosciurus erythraeus*	LC	LC	/	237–1077	E,M,S,V	1077	0.75	25.91
Red-hipped squirrel *Dremomys pyrrhomerus*	LC	NT	/	237–1098	E,M,S	988	0.57	23.77
Maritime striped squirrel *Tamiops maritimus*	LC	LC	/	310–1077	E,M,S	28	0.08	0.67

[a,b] CR, Critically Endangered; EN, Endangered; VU, Vulnerable; NT, Near Threatened; LC, Least Conscern; NE, Not Evaluated; [c] [illegible], China national first-class key protected wildlife; [illegible], China national second-class key protected wildlife; [d] E, tropical evergreen monsoon forest; M, tropical montane rainforest; S, tropical semdeciduous monsoon forest; V, tropical valley rainforest

Mammalian distribution in different habitat types of Jianfengling

From the camera-trap data on species distribution, the highest mammal abundance was observed in the tropical evergreen monsoon forests, followed by the tropical montane rainforests, with the lowest recorded in the tropical valley rainforests. Except unidentified rat, three species, namely rhesus monkey (*Macaca mulatta*), wild boar (*Sus scrofa*), and pallas's squirrel (*Callosciurus erythraeus*) were found across all the habitat types. By contrast, 13 other species only occurred in some specific habitats (Table 1; Fig. 3). There were five species with GO values greater than 0.50, namely wild boar, pallas's squirrel, Asiatic brush-tailed porcupine (*Atherurus macrourus*), red-hipped squirrel (*Dremomys pyrrhomerus*), and small-toothed ferret badger (*Melogale moschata*). There were five species with GO values less than 0.10, namely the Chinese pangolin, black giant squirrel, yellow-bellied weasel (*Mustela kathiah*), Malayan porcupine (*Hystrix brachyuran*), and swinhoe's striped squirrel (*Tamiops swinhoei*)(Table 1).

Mammalian relative abundance in Jianfengling

The most prevalent species, indicated by the total number of independent capture events and the relative abundance within the entire monitoring area, was Asiatic brush-tailed porcupine (Fig. 3). In terms of the RAI captured by camera traps, the second most abundant species was wild boar, with an RAI of 27.62, followed by pallas's squirrel, with an RAI of 25.91, and Hainan muntjac, with an RAI of 24.70. Conversely, the species with the lowest relative abundance recorded with no more than fifteen independent capture events were Chinese pangolin, black giant squirrel, yellow-bellied weasel, and Malayan porcupine, each having an RAI value of less than 0.30 (Fig. 3). Additionally, the monitoring also captured a significant number of domestic dogs, as well as human disturbances such as tea picking and medicinal herb collection, even hunting, all of which had a combined RAI of 4.50.

Discussion

This study used a camera-trapping survey to identify 15 mammal species across 6 orders and 10 families within the Jianfengling area, including several rare and endangered species, such as Chinese pangolin, common palm civet, Hainan muntjac, and black giant squirrel. Among these detected species, one was marked as Critically Endangered on the IUCN Red List, two were marked as Critically Endangered or Endangered on the Red List of China's vertebrates, and five were China's national key protected wildlife (Table 1). Compared with other areas within Hainan Tropical Rainforest National Park [32, 33], this finding highlights the diverse and intact mammal community in Jianfengling and demonstrates that Jianfengling holds significant potential for mammal conservation.

Notably, our research confirmed the persistent existence of wild Chinese pangolin populations in Jianfengling. As the principal habitat of *Manis pentadactyla pusilla* [34], the insular population experienced a precipitous decline due to poaching and habitat fragmentation, with sightings of live specimens becoming exceedingly rare by 2008 [35]. While community interviews in 2015 suggested its residual distribution within Hainan Tropical Rainforest National Park [36], tangible field evidence remained conspicuously absent [37]. A pivotal breakthrough occurred from 2015–2018, when camera-trapping surveys in Jianfengling's core zones successfully documented pangolin activities [22, 23]. Current investigations have revealed significant advancements: the GO of pangolin species in Jianfengling has reached 0.08, with 12 independent photographic records, including inaugural evidence from peripheral forest zones. These findings underscore the crucial conservation value of the population for maintaining genetic diversity, ensuring population stability, and facilitating the ecological restoration of this insular pangolin subspecies.

To our knowledge, phylogenetic analysis supports the recognition of Hainan muntjac as a distinct species within the genus *Muntiacus* [38]. This research addresses the knowledge gap regarding Hainan muntjac and sheds light on its conservation status and habitat requirements. Following Hainan Island's biogeographic isolation after the late Pleistocene, this distinctive evolutionary crucible fostered exceptional speciation and subspecific differentiation [39]. As a paradigmatic island ecosystem, the island not only serves as the type locality for *Paradoxurus hermaphroditus hainana* [40], but our current investigation has methodically documented this taxon's extensive spatial distribution in Jianfengling. Notably, *Macaca mulatta hainana* was shaped through prolonged adaptive evolution [41], demonstrating remarkable ecological adaptive capacities across Jianfengling's elevational spectrum—from valley rainforests to montane forests.

As the biogeographic nucleus of Hainan Island's tropical forest ecosystems, Jianfengling's superlative ecological milieu establishes it as a critical sanctuary for numerous rare and endangered species. Our camera-trapping data reveal that this area sustains emblematic examples of flagship species guilds, and six mammal taxa from the *Hainan Tropical Rainforest National Park Priority Conservation List* have been systematically documented [42], including the Chinese pangolin, common palm civet, Hainan muntjac, rhesus monkey, mainland leopard cat and black giant squirrel. Comparative analysis against existing biodiversity baseline data from other park sectors demonstrated that Jianfengling's mammalian community is highly rich, accentuating its exceptional role as an insular ecological refuge sustaining evolutionary distinctiveness [32, 33].

The Jianfengling investigation revealed profound historical shifts in the megafaunal assemblage. Compared with 20th-century biodiversity baselines, our survey failed to document critical species, including clouded leopards, the Asian black bear (*Ursus thibetanus*), the Hainan gibbon, Eld's deer, and the sambar deer (*Rusa unicolor*) [21]. These absent taxa predominantly occupy high trophic strata (Primates, Carnivora, large Artiodactyla), exhibiting spatiotemporal extinction gradients: Hainan gibbon and Eld's deer vanished from Jianfengling after the 1980s [43, 44], whereas clouded leopards and Asian black bears experienced island-wide extirpation after the 1990s [22, 23, 45]. This ecological void has precipitated marked trophic cascades: extant wild boars and Asiatic brush-tailed porcupines demonstrate extraordinary RAIs of 27.62 and 44.21, respectively, and their demographic explosions are potentially linked to apex predator functional extinction [46]. Comparative analyses have confirmed strong positive correlations between megafaunal population collapse and habitat fragmentation in East Asian tropical forests [47–50]. The structural disequilibrium within Jianfengling's extant fauna community epitomizes insular manifestations of Empty Forest Syndrome, revealing cascading consequences of defaunation in relict ecosystems [51].

The preservation mechanisms underpinning mammalian biodiversity in Jianfengling face formidable challenges from multidimensional anthropogenic stressors. Our research identifies three core pressure factors: poacher incursions (wielding hunting rifles), tourism intensification (peaking at 2,500 daily visitors), and community activities (notably free-ranging domestic dogs), collectively manifesting an RAI of 4.50. In particular, 31 camera-trap vandalism incidents were recorded during the study period; these incidents were spatiotemporally correlated with the detection of two armed intruders, underscoring the encroachment intensity of illegal wildlife trade networks into protected areas [36]. Historical ethnological research reveals how postban cultural inertia in Li ethnic hunting traditions synergizes with habitat homogenization from ecotourism development, forging complex human–wildlife conflict matrices [19]. The Tianchi artificial lake tourism hub (> 100,000 annual visitors) exemplifies cascading impacts: stream habitat fragmentation has precipitated local extinctions of habitat specialists, including sambar deer, the oriental small-clawed otter (*Aonyx cinerea*), Javan mongoose (*Herpestes javanicus*), and crab-eating mongoose (*Herpestes urva*); these patterns are corroborated by monitoring gaps in Mo et al.'s 2019–2021 datasets [22, 23]. Current conservation practices demand strategy reconfiguration through human activity gradient modeling. We propose Maxent-driven delineation of disturbance hotspots with three-tiered buffer zones insulating core tourism areas, coupled with IPBES-aligned community comanagement frameworks to reconcile biocultural tensions [52]. This integrative approach addresses both the material infrastructure of ecological degradation and the epistemic challenges of traditional practice preservation in anthropocene conservation governance.

In conclusion, Jianfengling harbors a diverse and unique mammal assemblage that includes diverse endangered and endemic species. This richness holds immense conservation value, not only for the region and Hainan Tropical Rainforest National Park but also for the global ecosystem. The presence of human disturbance or other potential threats underscores the urgency of immediate habitat conservation actions and conflict mitigation strategies. This research provides essential data and establishes solid scientific groundwork for developing and implementing effective conservation measures aimed at preserving the mammal fauna of Jianfengling, supporting the importance of conservation efforts in Jianfengling. Additionally, the importance of mammal conservation in Jianfengling will provide valuable insights for protecting mammal diversity and promoting effective wildlife conservation.

Declarations

Supplementary materials

Number of independent detections in the Jianfengling Branch of Hainan Tropical Rainforest National Park, China.

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Author contributions

W.Y. conceived of the research program, acquired funding, supervised all elements of the research, and wrote the manuscript. Z.Z. conducted conceived of the research program, helped write and edit the manuscript. S.X., S.M., C.L, W.Y., Z.H., L.L., and Z.L. conducted field data collection. Q.W. conducted field data analyses and helped review and edit the manuscript. All authors have read and agreed to the published version of the manuscript.

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Data availability statement

The data presented in this study are available upon request from the corresponding author.

Institutional review board statement

Ethical review and approval were waived for this study because infrared camera traps facilitate valuable data without disturbing wildlife or compromising ethical considerations.

Informed consent statement

Not applicable.

Conflicts of interest

The authors have no relevant financial or nonfinancial interests to disclose.

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Figures

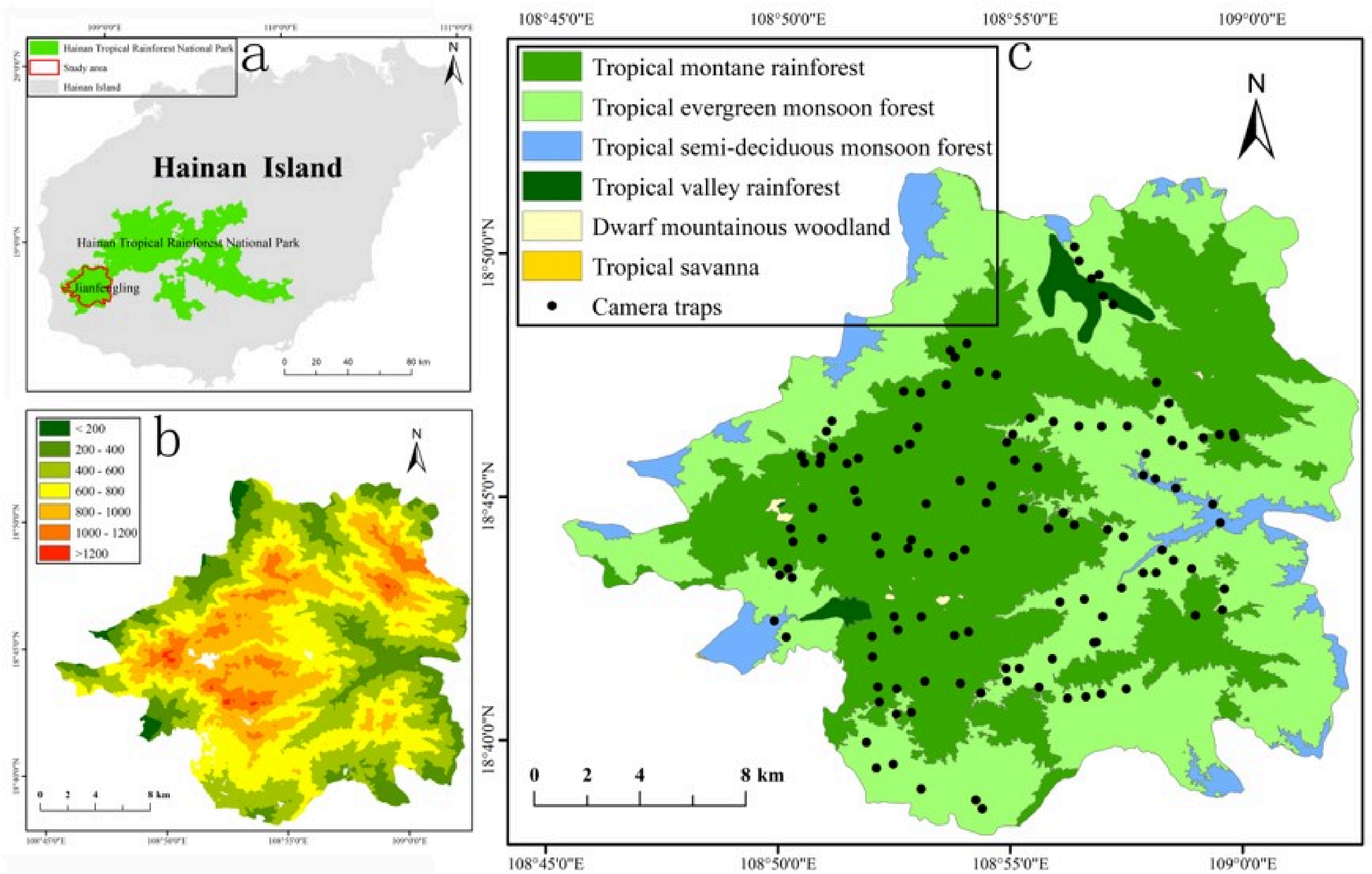

Figure 1

(a) Study area and Hainan Tropical Rainforest National Park, China. (b) Digital elevation model (DEM) derived from 1:1 million data provided by the National Catalog Service for Geographic Information (https://www.webmap.cn/main.do?method=index, accessed on 10 October 2024), and (c) vegetation type derived from 1:100 thousand data provided by the Jianfengling Branch Bureau of Hainan Tropical Rainforest National Park Management Office. The inset black dots represent the locations of the camera traps.

Figure 2

Infrared camera-trap images of the Chinese pangolin (*Manis pentadactyla*), Hainan muntjac (*Muntiacus nigripes*), black giant squirrel (*Ratufa bicolor*), common palm civet (*Paradoxurus hermaphrodites*), rhesus monkey (*Macaca mulatta*), and mainland leopard cat (*Prionailurus bengalensis*).

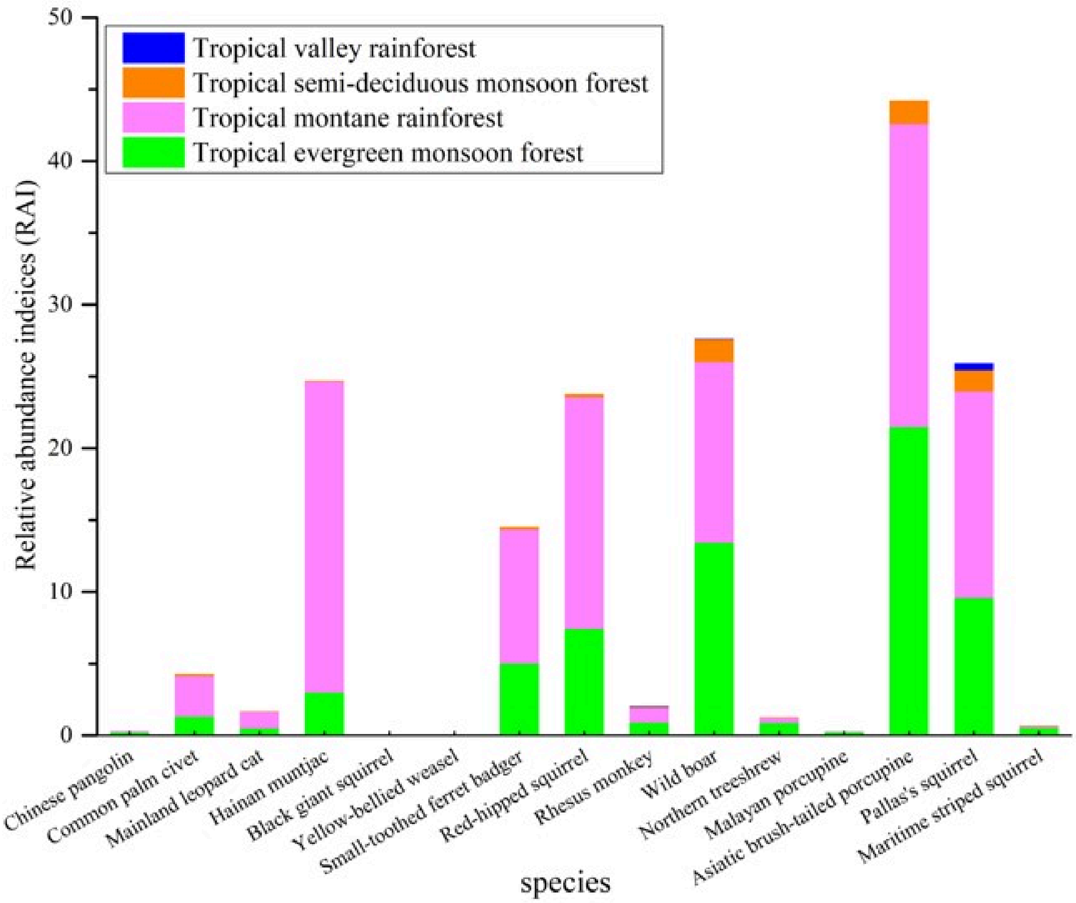

Figure 3

The relative abundance indices (RAIs) of species are depicted, with different colors in the columns representing the percentage of traps recording species across various habitat types in the Jianfengling Branch of Hainan Tropical Rainforest National Park, China.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- NumberofindependentdetectionsinJianfengling.xlsx